Notes on Random Communication Protocols

1 Randomized Protocols

1.1 Basics

Public randomness: all parties share a random string R. Private randomness: each party samples its own random string. Fixing randomness yields a deterministic protocol, so a randomized protocol is a distribution over deterministic ones. Error can be worst-case or with respect to an input distribution. Randomness often trades a small error for fewer bits.

Lemma 1.1. *Every private coin protocol can be simulated by a public coin protocol.*

Lemma 1.2. *Every public coin protocol can be simulated by a private coin protocol, with a small increase in communication.*

Lemma 1.3. *If a randomized protocol never errors, then fixing the randomness yields a deterministic protocol that is always correct.*

1.2 Applications and Examples

1. Equality: given $x, y \in \{0,1\}^n$, decide if $x = y$. Public hash $h : \{0,1\}^n \to \{0,1\}^k$, send $h(x)$, error $\leq 2^{-k}$.

2. Equality via codes: given $x, y \in \{0,1\}^n$, decide if $x = y$. Use a code $C : \{0,1\}^n \to \Sigma^m$, send a random coordinate $(i, C(x)_i)$.

3. Greater-than: given $x, y \in [n]$, decide if $x > y$. Randomized equality on prefixes lets you find the first differing bit with $O(\log \log n)$ rounds.

4. Disjointness: given $X, Y \subseteq [n]$, decide if $X \cap Y = \emptyset$. Random filters reduce set sizes while preserving intersection.

5. Hamming distance: given $x, y \in \{\pm 1\}^n$, estimate $\Delta(x, y) = |\{i : x_i \neq y_i\}|$ within $\pm m$ by sampling k coordinates.

6. Set membership: Alice has $S \subseteq [n]$, Bob has $i \in [n]$, decide if $i \in S$. Alice sends a hash sketch of S.

7. Gap-Hamming: given $x, y \in \{0,1\}^n$, decide if $\Delta(x, y) \leq n/2 - \sqrt{n}$ or $\geq n/2 + \sqrt{n}$. Random sampling separates the two cases.

8. Inner product mod 2: given $x, y \in \{0,1\}^n$, compute $\langle x, y \rangle$ mod 2. Use shared random masks and parity sketches.
9. Indexing: Alice has $x \in \{0,1\}^n$, Bob has $i \in [n]$, output x_i. Random fingerprints help verify short answers.
10. Approximate counting: given $x \in \{0,1\}^n$, estimate $|x|_1$ within $\pm\alpha n$ using random samples.
11. Distinct elements: Alice and Bob each have streams A, B, estimate $|A \cup B|$ via random hash buckets.
12. Sparse recovery: given $x, y \in \mathbb{R}^n$ with $x - y$ k-sparse, Bob estimates $x - y$ from random linear sketches.
13. Triangle detection: Alice and Bob have edge sets $E_A, E_B \subseteq \binom{[n]}{2}$, decide if $E_A \cup E_B$ contains a triangle by sampling edges.
14. Closest pair: given point sets $P, Q \subset \mathbb{R}^d$, estimate $\min_{p \in P, q \in Q} \|p - q\|_2$ using random projections.
15. ℓ_2 norm estimation: given $x, y \in \mathbb{R}^n$, estimate $\|x - y\|_2$ from random sign sketches.